```
In [4]: legitimate_urls.head(10)
        phishing_urls.head(10)
Out[4]:
```

	Domain	Having_@_symbol	Having_IP	Path	Prefix_suffix_separation	Protocol	Redirection_//_symbol	Sub_domains	URL_Length	ag
0	asesoresvelfit.com	0	0	/media /datacredito.co/	0	http	0	0	0	
1	caixa.com.br.fgtsagendesaqueconta.com	0	0	/consulta8523211 /principal.php	0	http	0	1	1	
2	hissoulreason.com	0	0	/js/homepage /home/	0	http	0	0	0	
3	unauthorizd.newebpage.com	0	0	/webapps/66fbf/	0	http	0	0	0	
4	133.130.103.10	0	1	/23/	0	http	0	2	0	
5	dj00.co.vu	1	0	/css/	0	http	0	0	2	
6	133.130.103.10	0	1	/21/logar/	0	http	0	2	0	
7	httpssicredi.esy.es	0	0	/servico/sicredi /validarclientes /mobi/index.php	0	http	0	2	2	
8	gamesaty.ga	0	0	/wp-content ///yh/en/	0	http	1	0	2	
9	luxuryupgradepro.com	0	0	/ymailNew /ymailNew/	0	http	0	0	0	

Figure 1: First few lines of the data set

```
In [2]: !pip3 install pandas

Collecting pandas
  Using cached https://files.pythonhosted.org/packages/c3/e2/00cacecafbab071c787019f00ad84ca3185952f6bb9bca9550ed8
3870d4d/pandas-1.1.5-cp36-cp36m-manylinux1_x86_64.whl
Collecting numpy>=1.15.4 (from pandas)
  Using cached https://files.pythonhosted.org/packages/45/b2/6c7545bb7a38754d63048c7696804a0d947328125d81bf12beaa6
92c3ae3/numpy-1.19.5-cp36-cp36m-manylinux1_x86_64.whl
Collecting python-dateutil>=2.7.3 (from pandas)
  Using cached https://files.pythonhosted.org/packages/36/7a/87837f39d0296e723bb9b62bbb257d0355c7f6128853c78955f57
342a56d/python_dateutil-2.8.2-py2.py3-none-any.whl
Collecting pytz>=2017.2 (from pandas)
  Using cached https://files.pythonhosted.org/packages/d3/e3/d9f046b5d1c94a3aeab15f1f867aa414f8ee9d196fae6865f1d6a
0ee1a0b/pytz-2021.3-py2.py3-none-any.whl
Collecting six>=1.5 (from python-dateutil>=2.7.3->pandas)
  Using cached https://files.pythonhosted.org/packages/d9/5a/e7c31adbe875f2abbb91bd84cf2dc52d792b5a01506781dbcf25c
91daf11/six-1.16.0-py2.py3-none-any.whl
Installing collected packages: numpy, six, python-dateutil, pytz, pandas
Successfully installed numpy-1.19.5 pandas-1.1.5 python-dateutil-2.8.2 pytz-2021.3 six-1.16.0

In [3]: !pip3 install sklearn

Collecting sklearn
Collecting scikit-learn (from sklearn)
  Using cached https://files.pythonhosted.org/packages/f5/ef/bcd79e8d59250d6e8478eb1290dc6e05be42b3be8a86e3954146a
dbc171a/scikit_learn-0.24.2-cp36-cp36m-manylinux1_x86_64.whl
Collecting scipy>=0.19.1 (from scikit-learn->sklearn)
  Using cached https://files.pythonhosted.org/packages/c8/89/63171228d5ced148f5ced50305c89e8576ffc695a90b58fe5bb60
2b910c2/scipy-1.5.4-cp36-cp36m-manylinux1_x86_64.whl
Collecting threadpoolctl>=2.0.0 (from scikit-learn->sklearn)
  Downloading https://files.pythonhosted.org/packages/61/cf/6e354304bcb9c6413c4e02a747b600061c21d38ba51e7e544ac7bc
66aecc/threadpoolctl-3.1.0-py3-none-any.whl
Collecting joblib>=0.11 (from scikit-learn->sklearn)
  Using cached https://files.pythonhosted.org/packages/3e/d5/0163eb0cfa0b673aa4fe1cd3ea9d8a81ea0f32e50807b0c295871
e4aab2e/joblib-1.1.0-py2.py3-none-any.whl
Collecting numpy>=1.13.3 (from scikit-learn->sklearn)
  Using cached https://files.pythonhosted.org/packages/45/b2/6c7545bb7a38754d63048c7696804a0d947328125d81bf12beaa6
92c3ae3/numpy-1.19.5-cp36-cp36m-manylinux1_x86_64.whl
Installing collected packages: numpy, scipy, threadpoolctl, joblib, scikit-learn, sklearn
Successfully installed joblib-1.1.0 numpy-1.19.5 scikit-learn-0.24.2 scipy-1.5.4 sklearn-0.0 threadpoolctl-3.1.0
```

Figure 2: Installing libraries

legitimate and the phishing ones, in order to make one data set. The first few lines of the merged data set are shown in Figure 2.

We then drop the columns like path, protocol, etc, that we do not need for the purpose of prediction:

```
urls = urls.drop(urls.
columns[[0,3,5]],axis=1)
```

After this, we need to split the data set into testing and training parts using the following code:

```
data_train, data_test, labels_train,
labels_test = train_test_split(urls_
without_labels, labels, test_size=0.30,
random_state=110)
```

We now make a model using the random forest classifier from sklearn, and then use the *fit* function to train the model:

```
random_forest_classifier =
RandomForestClassifier()
random_forest_classifier.
fit(data_train,labels_
train)
```

Once this is done, we can use the *predict* function to finally predict which URLs are phishing. The following line can be used for the prediction:

```
prediction_label = random_forest_
classifier.predict(test_data)
```

That is it! You have built a machine learning model that predicts if a URL is a phishing one. Do try it out. I am sure you will have fun. END

By: Jishnu Saurav Mittapalli

The author is a full stack Web developer, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.